

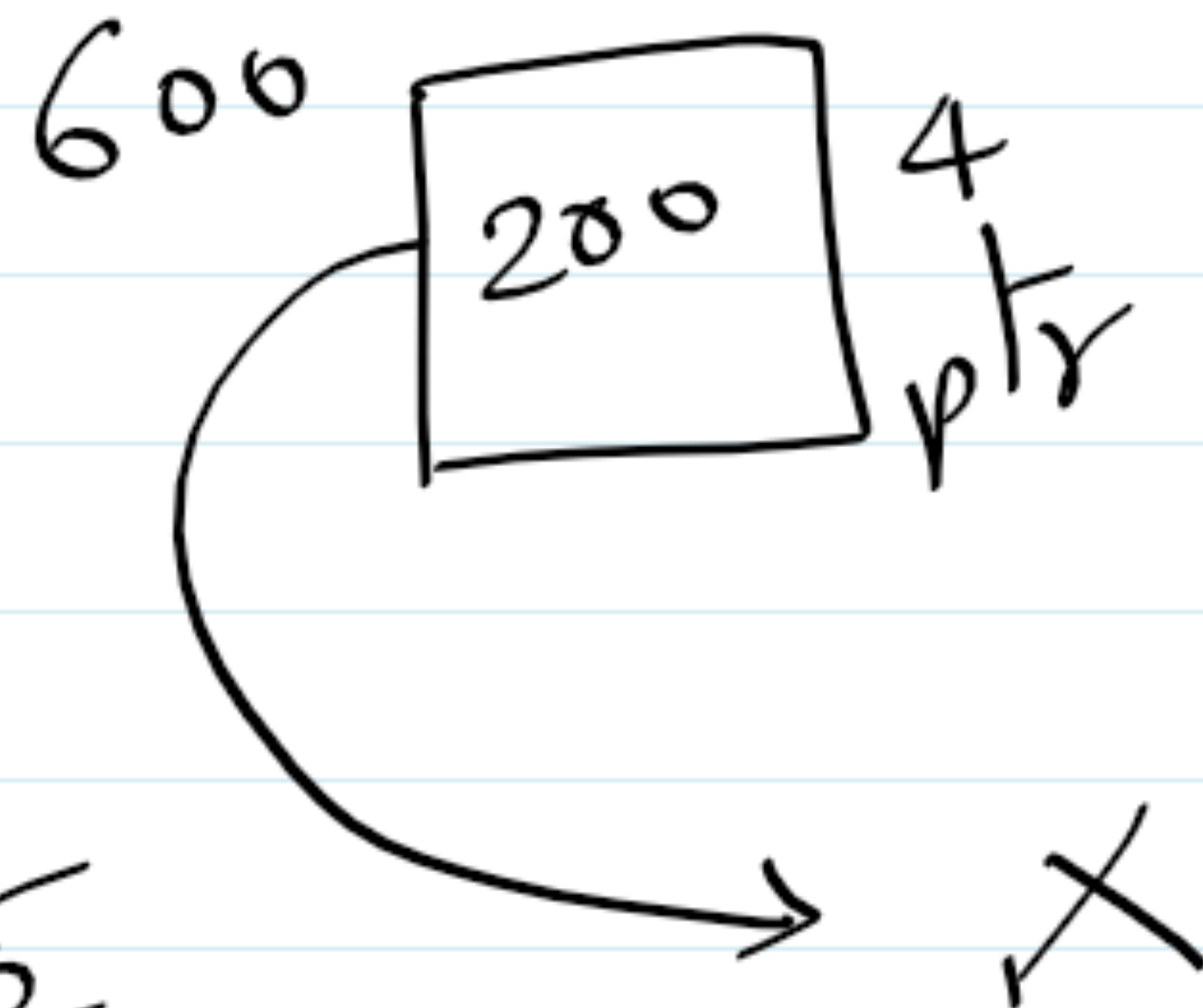
C++

✓ 200
10 value

```
void main()
{
    int value = 10;
    if (1)
    {
        int num = value;
    }
    num = 15;
    cout << value;
}
```

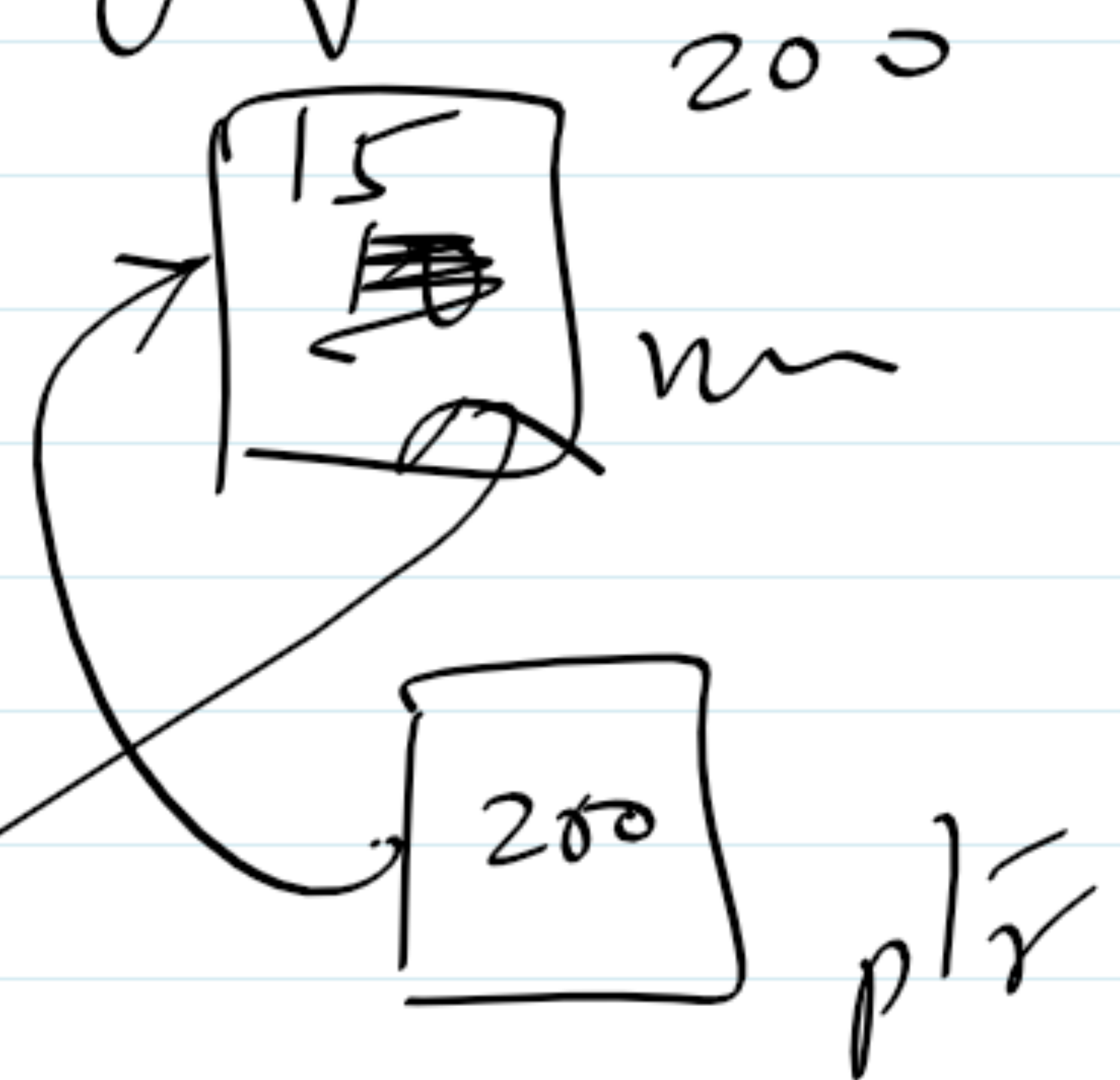


```
void main()
{
    int *ptr;
    if (1)
    {
        int num = 10;
        ptr = &num;
    }
    num = 15;
    cout << *ptr;
}
```



wild or dangling ptr

```
void main()
{
    int num = 10;
    if (1)
    {
        int *ptr = &num;
        *ptr = 15;
    }
    cout << num;
}
```



```
int *ptr;
void {
```

```
display
main {
int *ptr;
if (1) {
int *num = new int;
ptr = num;
*num = 15;
delete num;
ptr = null;
}
```

M2

→ 3
 → ~~delete ptr;~~
 → ~~ptr = null;~~
 → ~~ptr = null;~~

200
~~ptr~~
 ptr

1000
 mem

ptr = num;

① Dangling Pointer

X

X

② Memory Leak

1000

X

rr. Manage Memory Allocated

Managed Heap $\leftarrow$ JVM

② $\times$ delete $\rightarrow$ dangling pointer
 $\rightarrow$ $\times$ destructor

Garbage $\leftarrow$ Garbage Collector

Object

RC Table

MA	RC
1000	<u>0</u> <u>GC</u>

Data obj = new Data(); 1000

Data $\times$ null = obj;

Show(obj); u

p. v. Show(Data m)

~~obj = null;~~
OK

obj = xxy;
OK

~~obj~~

Memory Pool

500 bytes

✓	✓	✓	✓
✓	✓	✓	✓
✓	✓		
✓	✓	✓	

- ① OS → memory ↑ low
② GC → heap

Memory Management

GC → Destroy X

Object

finalize()

deprecated

→ P
GC

✓
Finalization

→ finalize GC

~~26~~ > = 6.0